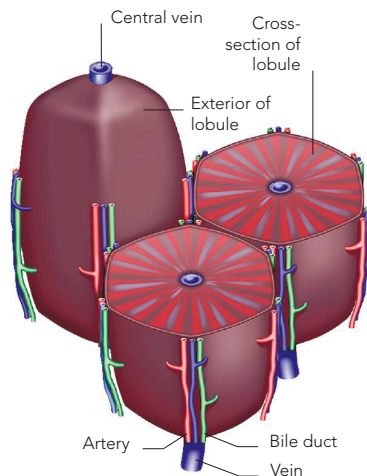


LIVER, GALLBLADDER, AND PANCREAS

THE LIVER IS THE LARGEST INTERNAL ORGAN AND MAKES, PROCESSES, AND STORES MANY IMPORTANT CHEMICALS. THE GALLBLADDER STORES BILE, AND THE PANCREAS SECRETES VITAL DIGESTIVE ENZYMES.

STRUCTURE AND FUNCTION OF THE LIVER

Weighing about 1.5kg (3 $\frac{1}{4}$ lb), the dark red liver is composed of lobules; these are made up of sheets of liver cells (hepatocytes), tiny branches of the hepatic artery and vein, and bile ducts. Nutrient-rich blood arrives from the intestines via the hepatic portal system (see p.222) and filters through the lobules. The liver has many functions, the most important of which are storing and releasing blood glucose for energy; sorting and processing vitamins and minerals; breaking down toxins into less harmful substances; and recycling old blood cells.



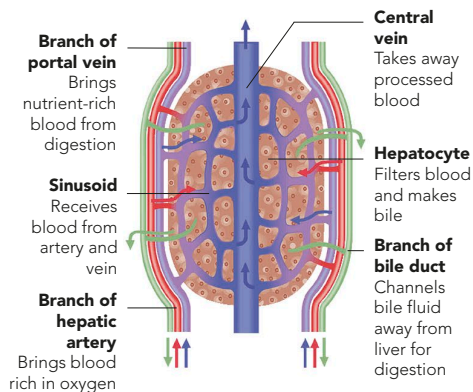
LIVER FUNCTIONS

Most of the liver's tasks are concerned with the process of metabolism. They include breaking down, storing, and circulating vital substances, and constructing complex molecules, such as enzymes.

BILE PRODUCTION	Secretes bile into ducts that lead to the gallbladder.
NUTRIENT PROCESSING	Converts sugars into glycogen and makes amino acids.
GLUCOSE REGULATION	Maintains the level of glucose in the blood.
DETOXIFICATION	Removes harmful substances such as alcohol from the blood.
PROTEIN SYNTHESIS	Makes blood-clotting proteins and proteins for blood plasma.
MINERAL AND VITAMIN STORAGE	Stores iron, copper, and vitamins.
BLOOD WASTE DISPOSAL	Eliminates bacteria and general foreign particles.
RECYCLING BLOOD CELLS	Breaks down red blood cells and reuses their constituents.

LIVER LOBULES

The six-sided lobules of the liver nestle together, each supplied with a central vein and blood vessels and bile-collecting vessels around them.



INSIDE A LOBULE

Hepatocytes filter the incoming, nutrient-rich blood into constituents that are destined for the bile ducts, storage, or waste disposal.